

Git & MLOps Assignment 1 Report

Version Control with Git, GitHub, and Visual Studio Code

Name: Aousaf Sulaman

Roll Number: 25L-8013

Course: MLOps

Assignment 1: Version Control with Git and GitHub

Field	Details
Assignment	Version Control with Git, GitHub, and Visual Studio Code
Deadline	9/9/2026 11:55 PM

Public GitHub Repository: <https://github.com/Aousaf90/mlops-project-25L-8013.git>

1. Introduction & Objective

This assignment builds a foundational MLOps workflow for a breast cancer classification project.

Student ID 25L-8013 is used in the repository name, branch names, training script filename, and commit messages.

2. Part 1 — Environment Configuration & Repository Initialization

2.1 Repository structure

The project uses the following layout:

```

├─ data/                                # Raw dataset (ignored by Git)
├─ src/
│   └─ train_25L8013.py                # Training script
├─ model/                              # Trained model output (ignored by Git)
├─ docs/
│   └─ Git_MLOps_Report_25L-8013.md
├─ .gitignore
├─ requirements.txt
└─ README.md

```

2.2 Model training script

src/train_25L8013.py performs:

1. Loads the dataset from data/
2. Trains a Scikit-Learn Logistic Regression model (with a preprocessing scaler in a pipeline)
3. Saves the trained model to model/ as a joblib file

2.3 Reproducibility

- requirements.txt lists the environment dependencies
- README.md documents install and run commands from the repository root

2.4 GitHub repository

- Repository name: mlops-project-25L-8013
- Remote URL: <https://github.com/Aousaf90/mlops-project-25L-8013.git>
- Default branch: main

```

aousaf@Aousafs-MacBook-Air Assignment_1_git % git remote -v
origin  https://github.com/Aousaf90/mlops-project-25L-8013.git (fetch)
origin  https://github.com/Aousaf90/mlops-project-25L-8013.git (push)
aousaf@Aousafs-MacBook-Air Assignment_1_git % git config --global user.name
Aousaf Sulaman
aousaf@Aousafs-MacBook-Air Assignment_1_git % git config --global user.email
59832341+Aousaf90@users.noreply.github.com
aousaf@Aousafs-MacBook-Air Assignment_1_git % git branch
feature-preprocessing-25L8013
feature-tuning-25L8013
* main
aousaf@Aousafs-MacBook-Air Assignment_1_git % git log
commit a900a67506cd562ed125eeab2e4585ff5b005e06 (HEAD -> main, origin/main, origin/feature-tuning-25L8013, origin/HEAD, feature-tuning-25L8013)
Author: Aousaf Sulaman <59832341+Aousaf90@users.noreply.github.com>
Date: Sat Sep 5 13:48:55 2026 +0500

    25L-8013: Switch feature scaling to RobustScaler

commit c83f6b6fdb38d666b7878bbf8e643f9df67ea2c6b
Author: Aousaf Sulaman <59832341+Aousaf90@users.noreply.github.com>
Date: Sat Sep 5 13:20:52 2026 +0500

    25L-8013: Add training pipeline, train model, and store artifacts in /data
aousaf@Aousafs-MacBook-Air Assignment_1_git %

```

3. Part 2 — Managing MLOps Artifacts with .gitignore

3.1 Purpose

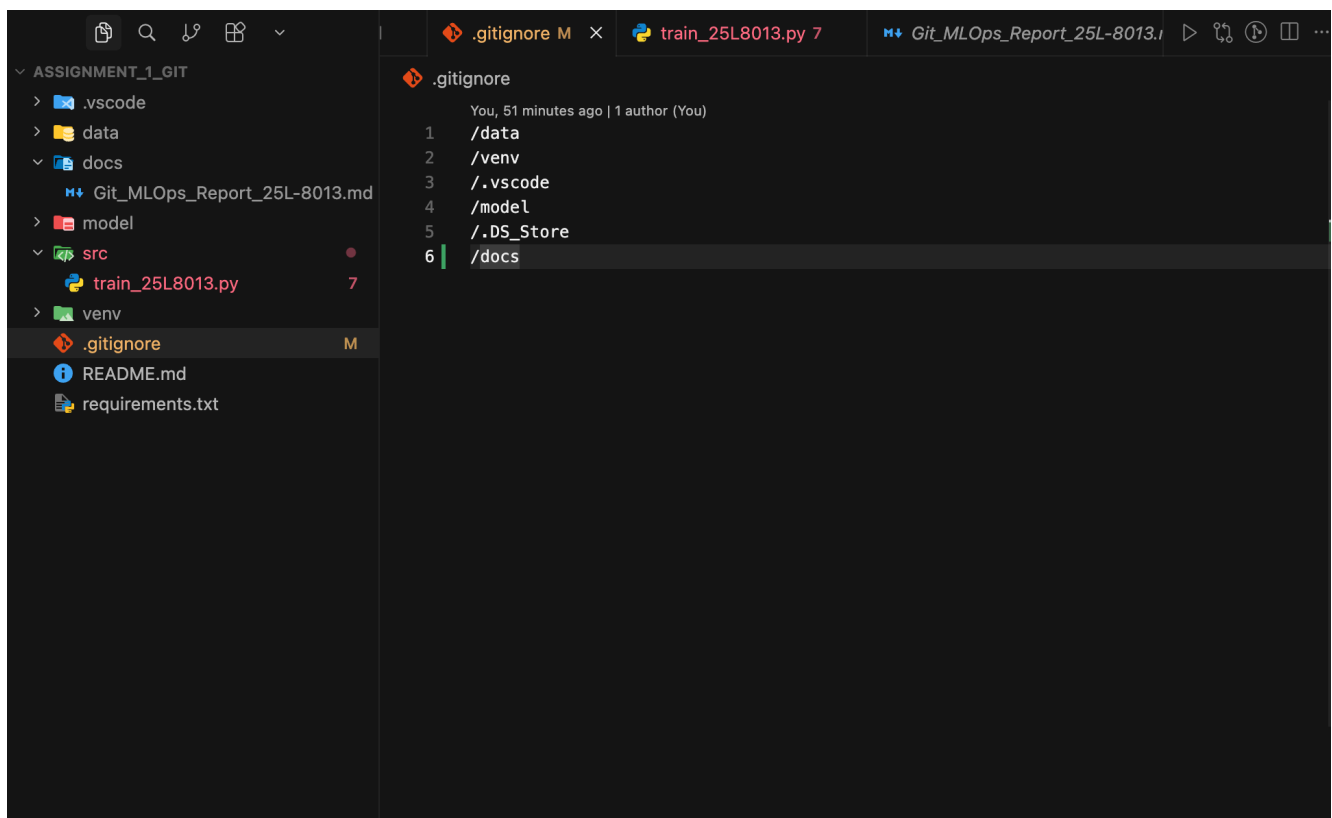
Raw datasets and heavy model binaries must not be pushed to GitHub. Only source code and configuration files are versioned.

3.2 .gitignore rules

The root .gitignore excludes:

- data/ contents (raw datasets)
- model/ contents (serialized models)
- Virtual environments (venv/)
- Editor / OS noise (e.g. .vscode/, .DS_Store)
- (Add any extra rules you configured, e.g. __pycache__/, *.pkl, *.h5, if present)

3.3 Verification



After placing data/model files locally, git status should not list them as untracked. Only code and config files should appear for commit.

Notes for this screenshot:

- Show the open .gitignore file in the editor
- Show Terminal or Source Control with git status where data/ and model/ artifacts do not appear as untracked

4. Part 3 — Staging, Snapshots, and Reset Operations

4.1 Iterative experimentation

A hyperparameter (or related training setting) was added/modified in `src/train_25L8013.py`. Uncommitted changes were inspected with VS Code Diff / git diff.

4.2 Soft and hard resets

Workflow practiced:

1. Modify the training script
2. Stage the change (git add)
3. Soft reset — unstage while keeping file edits (git reset / git reset --soft)
4. Hard reset — discard local changes back to last commit (git reset --hard) (use carefully)

```
laousaf@Aousafs-MacBook-Air Assignment_1_git % git status
On branch main
Changes not staged for commit:
  (use "git add <file>..." to update what will be committed)
  (use "git restore <file>..." to discard changes in working directory)
    modified:   .gitignore
    modified:   src/train_25L8013.py

no changes added to commit (use "git add" and/or "git commit -a")
laousaf@Aousafs-MacBook-Air Assignment_1_git % git diff src/train_25L8013.py
diff --git a/src/train_25L8013.py b/src/train_25L8013.py
index 5d094a7..5bc852b 100644
--- a/src/train_25L8013.py
+++ b/src/train_25L8013.py
@@ -44,15 +44,19 @@ X_train, X_test, y_train, y_test = train_test_split(
 # 4. Create Logistic Regression pipeline
# =====
+# Hyperparameter for experimentation (Part 3 - soft/hard reset demo)
+learning_rate = 0.01 # inverse of C; used to set regularization strength
+C = 1.0 / learning_rate
+
model = Pipeline([
    ("scaler", RobustScaler()), # median/IQR feature scaling
+    ("logistic_regression", LogisticRegression(
+        C=C,
+        max_iter=1000,
+        random_state=42
+    ))
])

-
# =====
# 5. Train model
# =====
laousaf@Aousafs-MacBook-Air Assignment_1_git % git add src/train_25L8013.py
laousaf@Aousafs-MacBook-Air Assignment_1_git % git status
On branch main
Changes to be committed:
  (use "git restore --staged <file>..." to unstage)
    modified:   src/train_25L8013.py

Changes not staged for commit:
  (use "git add <file>..." to update what will be committed)
  (use "git restore <file>..." to discard changes in working directory)
    modified:   .gitignore
```

```

|aousaf@Aousafs-MacBook-Air Assignment_1_git % git reset HEAD src/train_25L8013.py
Unstaged changes after reset:
M   .gitignore
M   src/train_25L8013.py
|aousaf@Aousafs-MacBook-Air Assignment_1_git % git status
On branch main
Changes not staged for commit:
  (use "git add <file>..." to update what will be committed)
  (use "git restore <file>..." to discard changes in working directory)
        modified:   .gitignore
        modified:   src/train_25L8013.py

no changes added to commit (use "git add" and/or "git commit -a")
|aousaf@Aousafs-MacBook-Air Assignment_1_git % git add src/train_25L8013.py
aousaf@Aousafs-MacBook-Air Assignment_1_git % git commit -m "25L-8013: temp hyperparameter for reset demo"

[main 803eaba] 25L-8013: temp hyperparameter for reset demo
1 file changed, 5 insertions(+), 1 deletion(-)
aousaf@Aousafs-MacBook-Air Assignment_1_git % git status

On branch main
Changes not staged for commit:
  (use "git add <file>..." to update what will be committed)
  (use "git restore <file>..." to discard changes in working directory)
        modified:   .gitignore

no changes added to commit (use "git add" and/or "git commit -a")
aousaf@Aousafs-MacBook-Air Assignment_1_git % git reset --soft HEAD~1

aousaf@Aousafs-MacBook-Air Assignment_1_git % git status

On branch main
Changes to be committed:
  (use "git restore --staged <file>..." to unstage)
        modified:   src/train_25L8013.py

Changes not staged for commit:
  (use "git add <file>..." to update what will be committed)
  (use "git restore <file>..." to discard changes in working directory)
        modified:   .gitignore

|aousaf@Aousafs-MacBook-Air Assignment_1_git % git reset --hard HEAD
HEAD is now at a900ae7 25L-8013: Switch feature scaling to RobustScaler
aousaf@Aousafs-MacBook-Air Assignment_1_git % git status

On branch main
Untracked files:
  (use "git add <file>..." to include in what will be committed)
        docs/

nothing added to commit but untracked files present (use "git add" to track)
aousaf@Aousafs-MacBook-Air Assignment_1_git % git diff src/train_25L8013.py

```

5. Part 4 — Branching, Collaboration, and Conflict Resolution

5.1 Feature branches

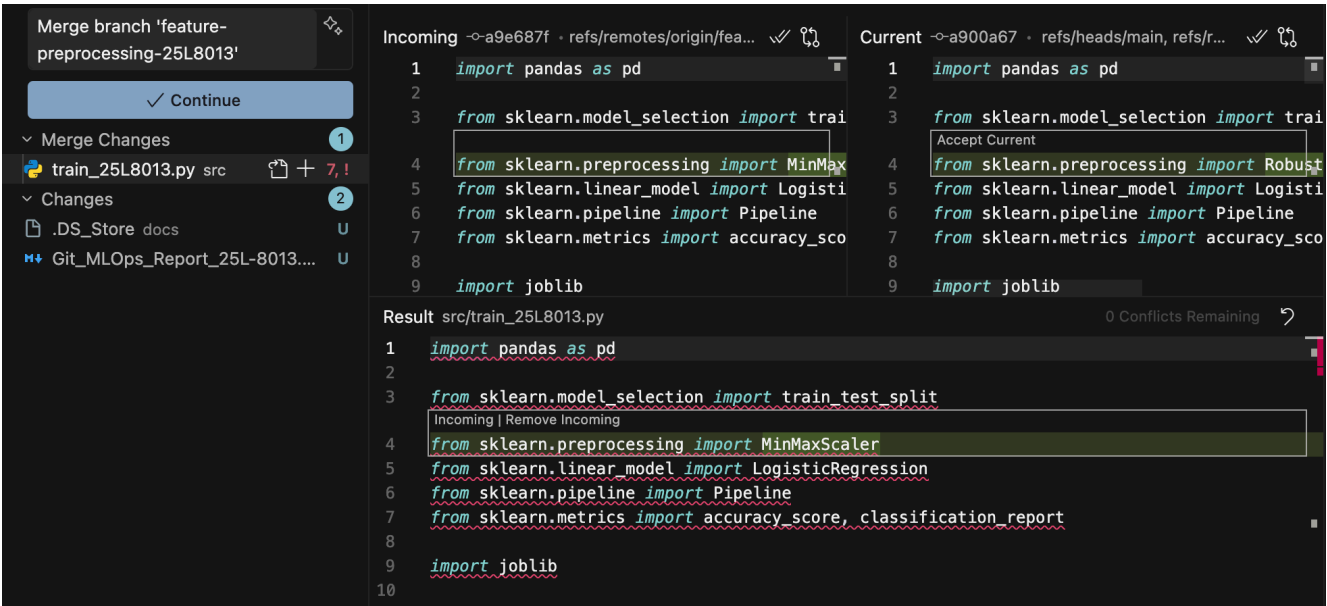
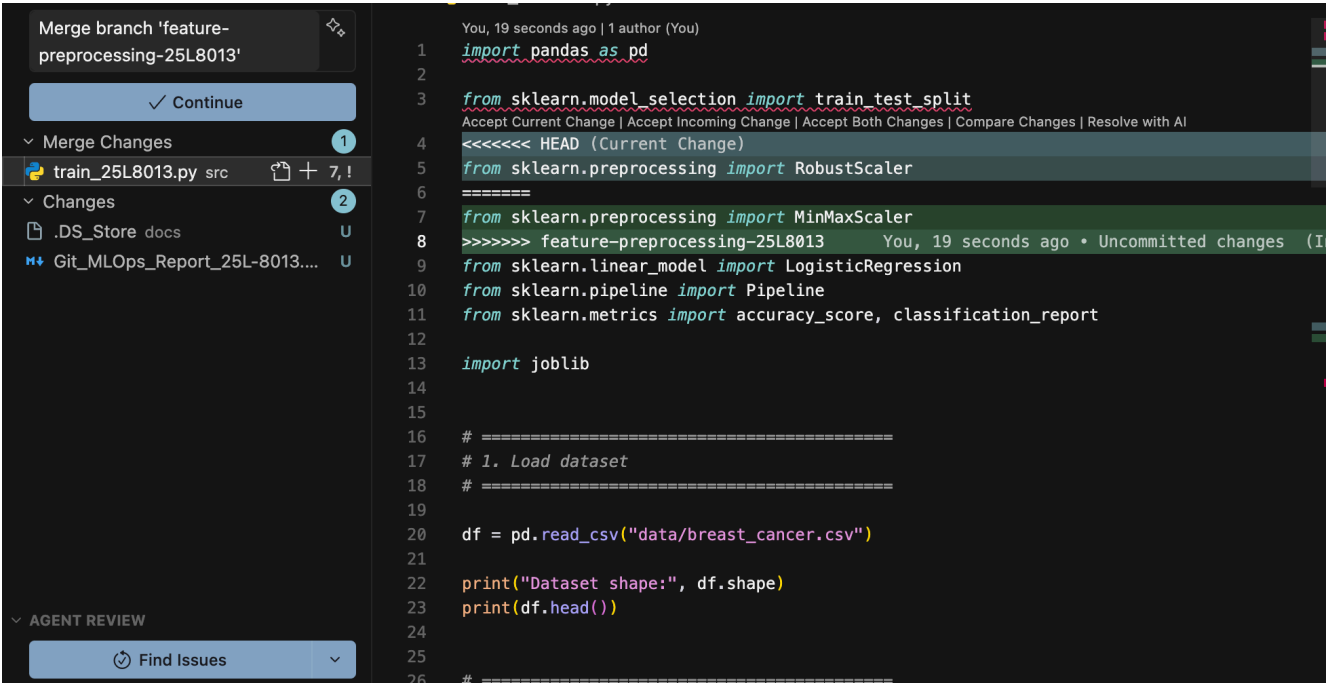
Branch	Purpose
feature-preprocessing-25L8013	Data normalization / preprocessing change
feature-tuning-25L8013	Alternative feature-scaling / tuning change

Both branches modified the same line in `src/train_25L8013.py` differently, which produced a merge conflict when integrating.

5.2 Conflict simulation and resolution

1. Create/push both feature branches
2. Merge one branch into main, then merge the second → conflict markers appear
3. Resolve in VS Code merge editor (<<<<<<<<, =====, >>>>>>>>)
4. Stage the resolved file, complete the merge commit, push main

Screenshot 3 (required) — VS Code merge conflict interface showing conflict markers



6. Part 5 — Inspection and History Management

6.1 Commit visualization

Commands used (or equivalent VS Code Git Graph / Timeline):

```
* 8b11453 (HEAD -> main, origin/main, origin/HEAD) 25L-8013: conflict merged
| \
| * a9e687f (origin/feature-preprocessing-25L8013, feature-preprocessing-25L8013) 25L-8013: Add MinMax data normalization step in training pipeline
| * a908a67 (origin/feature-tuning-25L8013, feature-tuning-25L8013) 25L-8013: Switch feature scaling to RobustScaler
| /
* c83f6bf 25L-8013: Add training pipeline, train model, and store artifacts in /data
aousaf@Aousafs-MacBook-Air Assignment_1_git %
```

6.2 Tracking file evolution

```
[aousaf@Aousafs-MacBook-Air Assignment_1_git % git log --follow -- src/train_25L8013.py
commit a908a67506cd562ed125eeab2e4585ff5b805e06 (origin/feature-tuning-25L8013, feature-tuning-25L8013)
Author: Aousaf Sulaman <59832341+Aousaf90@users.noreply.github.com>
Date: Sat Sep 5 13:48:55 2026 +0500

    25L-8013: Switch feature scaling to RobustScaler

commit a9e687f3e8c300cf44d7cbb36863d7aa98c2d04d (origin/feature-preprocessing-25L8013, feature-preprocessing-25L8013)
Author: Aousaf Sulaman <59832341+Aousaf90@users.noreply.github.com>
Date: Sat Sep 5 13:30:35 2026 +0500

    25L-8013: Add MinMax data normalization step in training pipeline

Co-authored-by: Cursor <cursoragent@cursor.com>

commit c83f6bfdb38d666b7878bbf8e643f9df67ea2c6b
Author: Aousaf Sulaman <59832341+Aousaf90@users.noreply.github.com>
Date: Sat Sep 5 13:20:52 2026 +0500

    25L-8013: Add training pipeline, train model, and store artifacts in /data
aousaf@Aousafs-MacBook-Air Assignment_1_git %
```

6.3 Context switching with stash

```
git stash
git stash list
git stash pop
```

Screenshot 4 (required) — Final commit history graph after successful merge

05_commit_graph.png — Commit history graph

Notes for this screenshot:

- Prefer `git log --oneline --graph --all` in the VS Code terminal, or a Git Graph extension view
- The graph should show feature branches and the merge into main

Screenshot placeholder — stash / follow (optional)

06_stash_or_follow.png — git stash or git log --follow

7.1 Public repository link

<https://github.com/Aousaf90/mlops-project-25L-8013>

